Bala Sreerangam

Cloud Platform Engineer · College Park, MD · Open to relocation & remote
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SUMMARY

Cloud Platform Engineer with 4 years operating production AWS at scale — 30+ accounts, 50+ environments, and a multi-tenant SaaS platform serving 20+ enterprise customers across aviation, retail, hospitality and telecommunications. Depth in Infrastructure as Code, CI/CD, zero-downtime release engineering, observability and cloud security. AWS Certified DevOps Engineer – Professional and Security – Specialty; completing an M.Eng. in Cloud Computing at the University of Maryland.

TECHNICAL SKILLS

Cloud: AWS — EC2, ECS/Fargate, Lambda, IAM, Organizations, STS, VPC, Route 53, S3, RDS, DynamoDB, ElastiCache, ECR, API Gateway, Step Functions, CloudWatch, EventBridge, SNS, SQS, KMS, Secrets Manager, CloudFormation & StackSets, Glue, Athena, CloudTrail, Config, GuardDuty, Security Hub, Cost Explorer, Transfer Family. GCP and Azure (foundational).

IaC & CI/CD: Terraform (reusable modules, version upgrades, remote state), Atlantis, CloudFormation StackSets, GitHub Actions, CircleCI, Buildkite, Jenkins, Docker, Amazon ECR, multi-stage builds, semantic versioning, blue-green and rolling deployments, Kubernetes.

Languages: Python (boto3), SQL, Bash, Go, JavaScript, TypeScript, Java, C.

Security & Observability: IAM least privilege, cross-account roles, WAF, Shield, Firewall Manager, GuardDuty, Security Hub, Config, CloudTrail, KMS, secret-leak scanning in CI, CIS AWS Foundations, Jamf; Datadog, CloudWatch, Grafana, monitor and threshold design, alert-noise reduction, on-call.

Data & Platform: AWS DMS, Glue, Airflow, Redshift, Athena, QuickSight, S3 data lakes, Snowflake, PostgreSQL, Redis, ETL design; multi-account AWS, landing zones, multi-tenant SaaS, microservices, serverless, event-driven architecture, high availability, disaster recovery, Chef, FinOps and rightsizing.

PROFESSIONAL EXPERIENCE

DevOps / Cloud Engineer — LJI

Jan 2022 – Jul 2025 · Multi-tenant enterprise loyalty platform · 30+ AWS accounts, 50+ environments, clients across multiple time zones. Joined as an intern in the final undergraduate semester and grew into owning release engineering, observability and infrastructure automation.

- Introduced **Blue/Green RDS deployments**, cutting deployment time approximately **80%**; combined with pre/post-migration sequencing on a secondary ECS task to achieve **zero-downtime releases** for any release without an engine upgrade.
- Rebuilt Datadog monitoring from **50–60 pages per day at roughly 10% true positives down to 10–15 actionable alerts — a 75–80% reduction in alert volume** — by setting per-environment and per-microservice thresholds from observed traffic, restoring on-call trust across the team.
- Reduced CI security scanning and release tag creation time **60–70%** by parallelizing scan jobs, and added secret-leak detection to catch credentials before a tag was cut.
- Planned and executed **30+ production deployments per week**, including major version upgrades and database migrations, sizing each window from release notes, lower-environment dry runs and Datadog traffic curves.
- Eliminated developer environment drift by building a **self-service deployment endpoint** on AWS Lambda and API Gateway backed by a Python/boto3 service map, so developers deployed by submitting service, environment and tag rather than hand-modifying ECS, Lambda and RDS configuration.
- Automated multi-step **tenant cloning with Step Functions** across RDS, S3 and DynamoDB, plus DMS-to-S3 and Airflow DAG resync, replacing a long manual runbook for the 5–6 clones QA and demo environments required weekly.
- Built an **organization-wide audit collector** in Python covering 50+ AWS services in every account, writing to a hierarchical S3 layout in a central account, queried through Athena with email alerting on findings requiring immediate action.
- Refactored the Terraform estate into **reusable DRY modules** spanning EC2, ECS, Lambda, VPC, RDS, DynamoDB, S3, CloudFront, API Gateway and load balancers, and co-designed a module version upgrade path so live environments could advance without being rebuilt.
- Owned the reporting data pipeline: AWS DMS full-load and ongoing replication from RDS, structured S3 staging, Airflow loads into Redshift, then Athena and QuickSight with role-scoped dashboard access for BI, audits and client reporting.
- Automated CloudFront provisioning for client websites, migrated AWS Transfer SFTP servers from hostname to static IP with no file loss, and hardened production with **WAF, Shield, Firewall Manager, Security Hub and GuardDuty** — including WAF web ACLs enforcing client-supplied IP allowlists on CloudFront — and implemented **Atlantis** so Terraform

plan and apply ran through pull requests, creating an auditable trail for every infrastructure change.

Graduate Teaching Assistant — University of Maryland

Aug 2026 – Present · MSML 650, Cloud Computing

- Guide graduate students through hands-on labs on AWS infrastructure, virtualization, containerization, cloud networking, storage, identity management and scalable application deployment.
- Hold weekly office hours, troubleshoot student cloud deployments, grade technical assignments, and mentor on cloud-native development, distributed systems and infrastructure automation.
- Collaborate with the instructor to improve lab exercises and evaluate cloud projects built in production-style environments.

Cloud Infrastructure & DevOps Engineer — Quanticore (My Behavior Bridge)

May 2026 – Aug 2026

- Built the first **Infrastructure as Code** in the organization using reusable Terraform modules covering VPC networking, subnets, route tables, NAT and internet gateways, security groups, IAM, EC2, ECS, load balancing, auto scaling, ECR and DNS, eliminating manual console provisioning.
- Designed the **environment strategy**, proposing and standing up a dedicated QA tier between development and production to give releases end-to-end validation before customer exposure.
- Built the **first CI/CD pipelines in GitHub Actions** — source-triggered workflows, security checks during build, automated Docker image creation, ECR publishing and release tags — then automated tenant provisioning and established the monitoring, networking and IAM practices the platform runs on.

Student Staff — University of Maryland, Accessibility & Disability Service

Feb 2026 – May 2026 · Part-time

- Coordinated accommodated exam operations and scheduling workflows, and built a reusable web-scraping tool for instructor email collection plus a Google Sheets scheduling application with constraint-based assignment logic and manual override.

PLATFORM PROJECTS

- **CloudForge** — Landing-zone platform provisioning a complete AWS environment from scratch in under 20 minutes, with a validate/plan/approve/deploy pipeline and remote state locking. *Terraform, Terragrunt, GitHub Actions, VPC, IAM, Route 53, ACM.*
- **AutoOps** — CI/CD platform carrying source to production with no manual step: security scanning, multi-stage Docker builds, semantic versioning, ECR publishing, rollback and post-deploy validation. *GitHub Actions, Docker, ECR, ECS, Terraform, Python.*
- **Sentinel** / **InfraGuard** — Event-driven governance platform that continuously scans QA, UAT and production for configuration drift and security issues across IAM, public exposure, encryption and logging, auto-remediates critical findings and centralizes reporting. *Lambda, EventBridge, SNS, CloudTrail, AWS Config, Python.*

RESEARCH

Stratosyn — Intelligent Cloud Governance & Autonomous Infrastructure Platform

Faculty-advised research · Sep 2025 – Present · University of Maryland

- Built a multi-account discovery engine using AWS Organizations enumeration, cross-account role assumption and concurrent multi-region scanning, solving sparse-schema normalization across heterogeneous boto3 responses — the schema for a service becomes the union of every key observed anywhere in the organization, null-filled where a resource omits it, stored as partitioned NDJSON in S3 with Glue and Athena providing schema-on-read. Researching automated compliance validation against CIS AWS Foundations, NIST and SOC 2, blast-radius graph generation, and safe remediation with rollback and audit trails.

CERTIFICATIONS

AWS Certified DevOps Engineer – Professional · **AWS Certified Security – Specialty** (valid through 2029) · AWS Certified Solutions Architect – Associate · AWS Certified Cloud Practitioner · HashiCorp Certified: Terraform Associate (003) · Jamf Certified Associate

EDUCATION

Master of Engineering (M.Eng.), Cloud Computing — University of Maryland

Sep 2025 – May 2027 (expected)